

DDPM Exam 2 — Comprehensive Cheat Sheet (Modules 4 & 5)

This document provides a comprehensive, source-faithful module-by-module review across all 15 syllabus topics for Exam 2 of the Data-Driven Product Management programme.

Operational Parameters & Scope Status

- **Confirmed Operational Parameters:** 45 questions | 60-minute duration (3:30 PM–4:30 PM IST, Sun 16 Aug) | No negative marking (answer 100% of questions).

- **Material:** Topics with missing local class assets are marked as **externally supplemented; instructor-specific coverage unresolved**.

Module 4: Building Product Market Strategy

Topic M04-T01: Marketing Strategy and Marketing Plan

1. Strategic Context & Expansion Options (Natureview Farm Case)

- **Option 1: 8 oz SKU Expansion into Supermarkets (East & Central)**
 - Target: Enter conventional supermarket channel with 6 SKUs of 8 oz yogurt.
 - Trade Promotions: 15% off-invoice promotion + co-op advertising allowances.
 - Broker Commission: 5% of manufacturer sales.
- **Option 2: 32 oz SKU National Expansion into Supermarkets**
 - Target: Enter conventional supermarket channel with 4 SKUs of 32 oz yogurt.
- **Option 3: 4 oz Multipack in Natural Foods Channel**

2. Baseline Financials (1999 Income Statement, Exhibit 1)

- Revenues: 13,000,000 (100%)
- Cost of Goods Sold (COGS): 8,190,000 (63%)
- Gross Profit: 4,810,000 (37%)
- Admin & Freight: 2,210,000 (17%)
- Sales Expenses: 1,560,000 (12%)
- Marketing Expenses: 390,000 (3%)

3. Key Formulas

Manufacturer Net Price = Retail Price $\times$ (1 - Retail Margin %) $\times$ (1 - Wholesaler Margin %)

Break-Even Volume (Units) = (Fixed Marketing Expenses (Slotting Fees + Ad Spend)) / (Manufacturer Contribution Margin per Unit)

Topic M04-T02: Lead Generation and Decoding Customer Journey

1. SERVQUAL Model (5 Service Quality Dimensions - RATER)

- **Reliability:** Ability to perform promised service dependably and accurately.
- **Assurance:** Employee knowledge, courtesy, and ability to inspire trust and confidence.
- **Tangibles:** Physical facilities, equipment, staff appearance, and digital UI touchpoints.
- **Empathy:** Caring, individualized attention provided to customers.
- **Responsiveness:** Willingness to help customers and provide prompt service.

2. Commitment-Trust Theory of Relationship Marketing

- **Trust:** Confidence in an exchange partner's reliability and integrity. Driven by shared values, communication, and non-opportunistic behavior.
- **Commitment:** Enduring desire to maintain a valued relationship. Reduces termination costs and increases customer retention.

3. Mental Accounting & Consumer Choice (Thaler)

- **Acquisition Utility:** $U_{acq} = v(p) - p$ (Value of goods received vs outlay).
- **Transaction Utility:** $U_{trans} = v(p^*) - p$ (Perceived deal quality relative to reference price p^*).
- **Four Principles of Mental Accounting:**
 - Segregate gains (multiple small pleasant surprises).
 - Integrate losses (single lump sum bill).
 - Integrate small losses with larger gains.
 - Segregate small gains from large losses (silver lining effect).

Topic M04-T03: Concept Testing and B2C Product Development (/)

1. Concept Testing Metrics & 5-Point Purchase Intent Scale

- Scale: 5 = Definitely Would Buy, 4 = Probably Would Buy, 3 = Might/Might Not Buy, 2 = Probably Would Not Buy, 1 = Definitely Would Not Buy.
- **Top 2 Box (T2B)** = % Definitely Buy + % Probably Buy.

2. Purchase Intent Discounting (Nielsen BASES Model)

Survey purchase intent overstates actual market behavior. Standard calibration discounting rules apply:

Calibrated Trial % = $(0.80 \times \text{Top Box \%}) + (0.30 \times \text{Second Box \%})$

3. ATAR Adoption Model (Urban & Hauser, 1993)

Forecasts market penetration and sales volume for new B2C products:

- **Awareness (A):** % of target market aware of the product.
- **Trial (T):** % of aware consumers making an initial trial purchase.
- **Availability (A):** % distribution coverage (ACV - All Commodity Volume).
- **Repeat (R):** % of trial buyers making repeat purchases.

Market Share % = $A \times T \times A \times R$

Topic M04-T04: Media Mix, Messaging, Packaging, and Bundling (/)

1. Integrated Marketing Communications (IMC) & Media Classification

- **Above-The-Line (ATL):** Mass media (TV, Radio, Print) to build mass awareness and brand equity.
- **Below-The-Line (BTL):** Direct promotions, point-of-sale displays, trade allowances driving immediate conversion.
- **Through-The-Line (TTL):** Integrated digital marketing, social media, search, and influencer campaigns.

2. Strategic Functions of Packaging (Kotler & Keller, 2015; Robertson, 2013)

- **Functional & Protective:** Preserves product integrity, safety, and shelf-life during logistics.
- **Informational & Regulatory:** Displays ingredient statements, nutritional facts, usage instructions, compliance labels, and barcodes.

3. Product Bundling Strategies (Stremersch & Tellis, 2002)

- **Pure Bundling:** Products sold exclusively as a bundle; components cannot be purchased individually.

- **Mixed Bundling:** Products offered both individually and as a discounted bundle.
- **Economic Rationale:** Captures consumer surplus across heterogeneous valuation segments.

Topic M04-T05: Managing Channel Partners and Affiliates (— externally supplemented; instructor-specific coverage unresolved)

1. Marketing Channel Architecture & Power (Coughlan et al., 2006)

- **Channel Conflict:**
- **Vertical Conflict:** Conflict between different channel levels (e.g., manufacturer selling D2C cheaper than retail partner MSRP).
- **Horizontal Conflict:** Conflict between channel members at the same level (e.g., price war between two regional distributors).
- **French & Raven's 5 Bases of Channel Power:** Coercive, Reward, Legitimate, Expert, Referent power.

2. Affiliate Marketing Payment Models

- **Pay-Per-Click (PPC):** Affiliate compensated per traffic click.
- **Pay-Per-Lead (PPL):** Affiliate compensated per qualified user lead/registration.
- **Pay-Per-Sale (PPS):** Affiliate compensated per completed transaction.

3. Marketing Attribution Platform Specifications

- **Google Analytics 4 (GA4) :**
- Retired legacy rules-based models (First Touch, Linear, Time-Decay, rules-based Position-Based) in mid-2023.
- Supports 3 models: **Data-Driven Attribution (DDA)** (machine learning sequence analysis), **Paid and Organic Last Click**, and **Google Paid Channels Last Click**.
- **Adobe Marketo Measure (Enterprise B2B Multi-Touch) :**
- **U-Shaped:** Allocates **50%** credit to First Touch, **50%** to Lead Creation Touch, and **0%** to intermediary touches.
- **W-Shaped:** Allocates **30%** to First Touch, **30%** to Lead Creation Touch, **30%** to Opportunity Creation Touch, and **10%** split equally across middle touches.

Topic M04-T06: Pricing Models

1. Strategic Pricing Paradigms

- **Cost-Plus Pricing:** Fixed markup added to cost; simple but ignores customer WTP.
- **Value-Based Pricing:** Prices set based on customer Willingness-to-Pay (WTP) and Economic Value to Customer (EVC).
- **Dynamic Pricing:** Algorithmic price adjustments based on real-time demand, time, and inventory constraints.

2. Tiers of Price Discrimination

- **First-Degree (Personalized):** Charging each customer their exact maximum WTP.
- **Third-Degree (Segmented):** Charging different prices to distinct customer segments (e.g., stadium grandstand vs bleacher seating).

3. Springfield Nor'easters Case Analysis

- **Optimal Price Formula:**

$$E_p = (\% \Delta Q) / (\% \Delta P), \text{ quad Optimal Price} = (MC) / (1 + (1 / E_p))$$

Module 5: Defining User Experiences - Tools

Topic M05-T01: AI in Customer Experience and Personalisation (— externally supplemented; instructor-specific coverage unresolved)

1. Personalization Architecture

- **Rule-Based vs AI Personalization:** Static demographic rules vs dynamic machine learning engines evaluating behavioral clickstreams and real-time context.
- **Next-Best-Action (NBA) Engine:** Algorithmic decisioning to serve the optimal offer, message, or support prompt in real time.

2. Recommendation Algorithms (Aggarwal, 2016)

- **Collaborative Filtering:**
 - User-Based: Recommends items preferred by users with similar rating histories.
 - Item-Based: Recommends items co-purchased or co-viewed with target item (e.g., "Customers who bought this also bought...").
- **Content-Based Filtering:** Recommends items matching features of items previously preferred by user.
- **Hybrid Systems:** Combines collaborative and content-based filtering to overcome cold-start issues.

3. Conversational AI & RAG Support

- NLU intent recognition, automated ticket triage, and Retrieval-Augmented Generation (RAG) over enterprise knowledge bases.

Topic M05-T02: Customer Engagement and Retention Strategies (— externally supplemented; instructor-specific coverage unresolved)

1. Core Retention Metrics (Fader & Hardie, 2007)

Customer Churn Rate (CR) = (Customers Lost During Period) / (Customers at Start of Period)

Customer Retention Rate (CRR) = (End Customers - New Customers) / (Start Customers) × 100

Customer Lifetime Value (CLV) = (ARPU × Gross Margin %) / (Monthly Churn Rate)

Net Promoter Score (NPS) = % Promoters (9-10) - % Detractors (0-6)

2. Cohort Analysis & Retention Curves

- Tracking user cohorts over time (T_0, T_1, T_2, dots).
- Retention curve flattening horizontally indicates true Product-Market Fit (PMF); downward sloping curve to zero indicates lack of PMF.

3. RFM Customer Segmentation

- **Recency (R), Frequency (F), Monetary (M)** value.
- Clusters: Champions (High R, High F, High M), At-Risk (Low R, High F, High M — primary win-back targets), Hibernating (Low R, Low F, Low M).

Topic M05-T03: Cyber Risk Management and Trust (— externally supplemented; instructor-specific coverage unresolved)

1. NIST Cybersecurity Framework (NIST CSF v2.0 Core Functions) (: NIST CSF v2.0)]

NIST CSF v2.0 establishes 6 core functions:

- **Govern (GV):** Establishes organizational cybersecurity risk strategy, policies, roles, and supply chain risk oversight (newly added 6th function in CSF v2.0).
- **Identify (ID):** Asset management, risk assessment, operational system discovery.
- **Protect (PR):** Access control, identity management, staff training, data encryption.
- **Detect (DE):** Continuous monitoring, anomaly detection, intrusion detection.
- **Respond (RS):** Incident triage, containment, mitigation, forensics.

- **Recover (RC):** Restoration of services post-incident and resilience planning.

2. Privacy by Design (PbD) & Zero Trust Architecture (ZTA)

- **Privacy by Design (7 Principles, Cavoukian 2009):** Proactive not reactive, Privacy as default, Privacy embedded into design, Full functionality, End-to-end security, Visibility/transparency, Respect for user privacy.
- **Zero Trust Architecture (NIST SP 800-207):** "Never trust, always verify." Guiding principles: explicit verification, least privilege, assume breach posture.

Topic M05-T04: Tableau Session (— externally supplemented; instructor-specific coverage unresolved)

1. Tableau Data Architecture & Visual Encodings

- **Dimensions vs Measures:** Dimensions are categorical/qualitative (slice data); Measures are numerical/quantitative (aggregated via SUM, AVG).
- **Discrete vs Continuous:** Discrete (Blue pills) creates distinct headers; Continuous (Green pills) creates continuous axes.

2. Level of Detail (LOD) Expressions

- **FIXED LOD:** FIXED [Region] : SUM([Sales]) — Computes aggregate at specified dimension granularity, ignoring view dimensions. Evaluated before dimension filters.
- **INCLUDE LOD:** INCLUDE [Customer ID] : AVG([Sales]) — Computes aggregate including specified dimensions in addition to view dimensions. Evaluated after dimension filters.
- **EXCLUDE LOD:** EXCLUDE [Category] : SUM([Sales]) — Computes aggregate excluding specified view dimensions. Evaluated after dimension filters.

3. Tableau Order of Operations (Filter Pipeline Execution)

- Extract Filters → 2. Data Source Filters → 3. Context Filters → **4. FIXED LOD Calculations** → 5. Dimension Filters → **6. INCLUDE/EXCLUDE LOD Calculations** → 7. Measure Filters → 8. Table Calculations.